# Time Is the Baseline to Beat: Lead Time over a Timer as the Test for VLA Failure Monitors

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***Abstract*—On RoboCasa365, a failed vision-language-action (VLA) rollout is one that reaches its time limit. Calibrated at a 5% false-positive rate (FPR), a timer flags every one of 380 prospective failures, so recall cannot separate a learned monitor from elapsed time; only warning lead can. We alarm when *either* a policy-internal score or a timer crosses its threshold, calibrate the pair with a finite-sample FPR guarantee, and compare with a per-task timer P built from each task's success durations. On one VLA and 38 tasks, P nearly matches the SAFE failure score fused with a single timer, and on a fresh preregistered cohort of 760 rollouts SAFE adds 0.030 of the time limit to P (95% CI 0.007–0.093), below the margin of 0.05 we declared in advance. A sequence detector trained for early separation on the same features, Hide-and-Seek, passes the same registered test: it adds 0.071 (CI 0.022–0.157) at 0.061 FPR. Retrained on zero inputs, the same model matches its ROC-AUC but adds no lead; given only each rollout's first inference, it adds none either. Elapsed time explains how well monitors separate failures; the extra warning develops during the rollout. Stage 2 tests a second VLA and whether earlier alarms raise success when they trigger a retry.**

## I. Introduction

Runtime monitors for VLA policies read the policy's hidden state and predict eventual failure [8], or score how unusual that state is relative to successful rollouts [27, 19]. They are compared with each other, but not with what the robot knows for free: how long the attempt has been running.

In RoboCasa365, as in most simulated manipulation benchmarks, an episode ends at first success or at a time limit, and in our test cohort every one of 380 failed rollouts ran to its limit. A timer at 90% of the limit therefore flags all of them at 4.7% FPR. Against it, a monitor can add only one thing: an earlier alarm at the same false-alarm rate, which leaves time to stop, reset or ask for help. We make this lead time the test, with three contributions.

1) **A testable hypothesis and endpoint.** We define the warning lead of an alarm as the fraction of the time limit left when it fires (zero for a missed failure) and state four hypotheses with decision rules fixed before the confirmatory data were analysed (Sec. IV).
2) **Calibrated OR-fusion with a guarantee.** We alarm when either a policy-internal score or the timer crosses its threshold, and calibrate both thresholds along a nested path with a split-conformal rule that bounds the FPR on a new successful rollout by $\alpha$ in finite samples (Proposition 1).
3) **Exploratory and confirmatory evidence.** A per-task timer nearly matches SAFE fused with a single timer, and most failures are still recoverable at the limit (Sec. VI). On fresh preregistered cohorts SAFE adds 0.030 of the limit over the per-task timer, below our 0.05 margin, while Hide-and-Seek [17] adds 0.071, and controls show that its extra warning is not learned elapsed time (Sec. VII).

Elapsed time explains most of how well monitors *separate* failures; lead over a per-task timer shows which ones also warn *earlier*. On timeout-defined benchmarks, monitors should be compared with a per-task timer and evaluated on failures that end early.

## II. Related Work

**Failure detection from policy internals.** SAFE learns a failure score from a VLA's last-layer features and thresholds it with a functional conformal band [8]; we reuse its implementation. Follow-ups change the training data [18], labels [17, 14], features [28, 31] or policy [22]. None reports an elapsed-time detector. SAFE removes the duration shortcut by truncating rollouts to equal length and, on its real robot, by running every rollout to a fixed horizon [8]; a deployed monitor can do neither. Duration is a shortcut feature [7], but its causal form is legitimate online evidence.

**Action-space and success-only detectors.** STAC [1] and VLA-FAIL [20] compare overlapping action chunks, and ActProbe [9] feeds chunk consistency and magnitude to a task-conditioned LSTM, beating internal-feature baselines on RoboCasa. Our policy executes each chunk in full, so overlap signals are undefined. A task-conditioned recurrent detector can also learn how long each task takes, which our timers make explicit and our controls test (Sec. V). FAIL-Detect [27] and FIPER [19] score inputs, outputs or embeddings against successful rollouts only. FIPER alarms when two conformal scores *both* exceed their thresholds, which suppresses false alarms; it reports no elapsed-time baseline, although length separates most of its tasks (Sec. VI). We alarm when *either* crosses, because the policy score and the timer catch failures at different times, and calibrate the pair jointly.

**World-model verifiers and VLM monitors.** Check-VLA [12] is the closest work on our benchmark. It verifies each action chunk with an action-conditioned world model on RoboCasa365, at a conformal 5% rate of unnecessary interventions, and repairs it with the same VLA. Foresight [24] uses world-model latents. Neither compares against elapsed

time; either could be fused with a timer. LLM and VLM monitors [1, 13, 21, 30, 6] are slower and complementary.

**Time, calibration and intervention.** VLA confidence drifts over a task [32]. Our time-risk baseline is a per-task survival-style estimate [10, 5], and lead time at a fixed false-alarm budget is the accuracy–earliness trade-off of early time-series classification [26]. Split-conformal thresholds give finite-sample false-positive control under exchangeability [3]; we extend the construction to a two-signal OR rule (Sec. V-C). RESCUE [2] finds that good detection scores do not guarantee lower intervention cost, and CheckVLA and FLARE [29] couple detection to repair; an earlier alarm matters only if acting on it raises success.

## III. Problem Setting

**Time.** The simulator runs *control steps* at RoboCasa's default 20 Hz. The policy executes each 16-step action chunk in full, so *policy inferences* $t = 1, 2, \ldots$ (0.8 s each) are our clock, and a monitor decides only at an inference. Task $k$'s limit of $T_k$ control steps is $H_k = \lceil T_k/16 \rceil$ inferences (29 to 272, or 22.5 to 217.5 s). Every fraction of time ($t/H_k$, $c$, $\ell$) is a fraction of $H_k$.

**Monitoring.** A monitor watches a rollout of a known task $k$ and decides at each inference $t$ whether to alarm, using only the policy features so far, $t$, the task and $H_k$. The first crossing is the alarm $T$ (in inferences). FPR is the fraction of successful rollouts with any alarm. A rollout of length $L$ inferences ends at success or at $H_k$; for a failure ($L = H_k$) the *warning lead* is $\ell = (L - T)/H_k$, or 0 if missed, and an alarm at the final inference counts with $\ell = 0$. The mean lead $\bar{\ell}$ combines recall and earliness. Detectors are compared at operating points fixed on a calibration cohort under FPR $\leq \alpha = 0.05$. Confidence intervals come from a paired bootstrap that resamples tasks, then rollouts within each task (2,000 replicates).

**Scope.** Whether elapsed time is a strong baseline depends on the termination rule. Where successes stop at completion (RoboCasa, FIPER's tasks, deployment), it is. Where every rollout runs to one fixed horizon, as in SAFE's real-robot set (in its release, 98% of failures and $\geq 30$ successes per task stop at one step) or SimplerEnv, it is uninformative, and early-warning claims need failure-time annotations, which SAFE did not release.

## IV. Hypotheses

We test four hypotheses. H1 and H2 were evaluated on a fresh confirmatory cohort with the decision rules below, registered as an amendment before any timer or lead quantity was computed on that cohort, and registered again for a second score before that score was computed on it (Sec. VII); H3 and H4 are the Stage 2 extensions (Sec. VIII).

**H1 (lead time, primary).** At thresholds calibrated for FPR $\leq$ 0.05, OR-fusing a policy-internal score with the per-task timer (Sec. V), the strongest time-only monitor we know, increases the mean warning lead over that timer alone. *Supported* if the paired bootstrap 95% CI of $\Delta\bar{\ell}$ lies above zero, the point gain is at least 0.05 of the time limit, and the fused test FPR is at most 0.075 with a Wilson 95% upper bound at most 0.10. As a secondary check, the fused detector must also beat a timer whose fraction is tuned *on the test cohort* to the fused detector's test FPR, which favours the timer.

**H2 (the gain needs policy evidence).** Fusions that contain no policy-internal evidence (Gaussian noise, or SAFE scores resampled within task, each fused with the per-task timer) gain less than 0.05 of the time limit over the per-task timer.

**H3 (generality).** On a second VLA with a different architecture (RLDX-1, 20 tasks), every failure runs to the time limit, P leads the global timer, and a policy score fused with P gains lead over P. SAFE + P and Hide-and-Seek + P are co-primary: each is *supported* if its decision CI lies above zero (97.5%, Bonferroni over the two) under the FPR rule of Sec. VIII; there is no 0.05 margin.

**H4 (utility).** When an alarm triggers a reset and retry, the OR-fused monitor yields a higher task success rate than the timer within the same per-task time budget.[TODO: Must change to match E3; author's call. Proposal: within a budget of 1.25 $H_k$, reset and retry triggered by P yields a higher success rate than no monitor; SAFE + P against P is descriptive.]

Only H1 and H2 are confirmatory; every other interval in this paper is descriptive. We expected H1 to fail for SAFE (opened-data gain 0.007); it did, with a gain of 0.030 whose CI excludes zero. A second amendment then applied the same rules to Hide-and-Seek, which gained 0.083 at matched FPR on the opened cohort; it passed (Sec. VII).

## V. Method

### A. Signals

**Policy score.** At each inference we take the final-layer action-token hidden state of the policy's action head (1,024 dimensions). The SAFE MLP [8] maps it to a failure score $u_t^{(j)}$ for each of three training seeds $j$. With per-task location $m_k^{(j)}$ and scale $\sigma_k^{(j)}$ estimated on development rollouts, the policy score is

$$s_t = \frac{1}{3}\sum_{j=1}^{3} \frac{\max_{t' \leq t} u_{t'}^{(j)} - m_k^{(j)}}{\sigma_k^{(j)}}. \tag{1}$$

The running maximum makes $s_t$ monotone and does not change when a threshold is first crossed. Success-only alternatives (Mahalanobis [11], kNN [23], PCA-KMeans and RND [4]) are fitted on successful development rollouts and aggregated the same way.

**Sequence score and controls.** Hide-and-Seek [17] trains an LSTM on a 128-dimensional PCA of the same hidden state with rollout labels only: a margin loss pushes each failure's peak score above every success's, and an onset loss acts within failures. Its scores are normalized and aggregated like SAFE's. A recurrent model can learn how long each task takes, so we retrain it unchanged on (i) zero inputs, (ii) the features of a random same-task training rollout at the same inference, and (iii) the rollout's own first-inference features, repeated. They see (i) elapsed time, (ii) also the task's typical trajectory, and

(iii) also the episode's initial appearance. ActProbe [9] reduces to chunk magnitude here, as our chunks never overlap.

**Timers.** The global timer alarms once $t \geq c\,H_k$ for a fraction $c \in (0, 1]$ shared by all tasks; it uses only the task's time limit. The per-task timer uses $c_k = \min(1, \rho_k + \delta)$, where $\rho_k$ is the 95th percentile of task $k$'s development success durations as fractions of $H_k$ and the shared margin $\delta$ is set by Proposition 1 on calibration successes. It needs successful rollouts of each task, but no failures and no policy features. We fixed the 95th percentile after auditing the opened cohort, so we report 0.75, 0.90 and 1.0 (the longest success) as sensitivity.

**Time risk (baseline).** The survival-style alternative estimates $r_t = \Pr(\text{failure} \mid k, \text{still running at } t)$ from development rollouts: with $n_t$ of them still running at $t$ and $f_t$ of those failing, it fits an increasing isotonic regression in $t/H_k$ to $(f_t + 0.5)/(n_t + 1)$, weighted by $n_t$. Because every failure runs to $H_k$, $r_t$ is a task-specific timer; its smoothing keeps it below a high threshold in tasks with few development failures, which is why it misses failures the plain timer catches.

### B. OR-fusion

The fused monitor alarms at

$$T = \min\{\, t : s_t > \tau_s \text{ or } t \geq c\,H_k \,\}. \tag{2}$$

Neither signal is restricted in time: the policy score may alarm early, and the timer guarantees an alarm on every failure, at $c\,H_k$ at the latest. The lead gain over the timer comes only from failures on which the policy score crosses first.

### C. Calibration with a finite-sample FPR guarantee

A successful rollout $i$ of length $L_i$ (in inferences) raises a false alarm under $(\tau_s, c)$ if and only if $M_i = \max_{t \leq L_i} s_t > \tau_s$ or $\rho_i = L_i / H_{k_i} \geq c$, so its false alarm depends only on the pair $(M_i, \rho_i)$. Fix in advance a path $\gamma(\lambda) = (\tau_s(\lambda), c(\lambda))$, $\lambda \in [0, 1]$, with both coordinates continuous and nondecreasing, so that a larger $\lambda$ is more conservative and $\gamma(1)$ never alarms on a success. For each calibration success define

$$V_i = \min\{\lambda : M_i \leq \tau_s(\lambda) \text{ and } \rho_i < c(\lambda)\}, \tag{3}$$

the least conservative point on the path at which rollout $i$ stays silent. Because the path is monotone, rollout $i$ is silent at $\lambda$ exactly when $\lambda \geq V_i$.[1]

*Proposition 1:* Let $n$ calibration successes and a new successful rollout be exchangeable, and let the path $\gamma$ not depend on the calibration successes. Set $\hat{\lambda} = V_{(\lceil (n+1)(1-\alpha) \rceil)}$, the $\lceil (n+1)(1-\alpha) \rceil$-th smallest $V_i$ (or $\hat{\lambda} = 1$ if that rank exceeds $n$). Then the OR-fused monitor at $\gamma(\hat{\lambda})$ raises a false alarm on the new success with probability at most $\alpha$.

*Proof:* The new rollout alarms at $\hat{\lambda}$ if and only if $V_{n+1} > \hat{\lambda} = V_{(r)}$ with $r = \lceil (n+1)(1-\alpha) \rceil$. By exchangeability, the rank of $V_{n+1}$ among $V_1, \ldots, V_{n+1}$ is uniform on $\{1, \ldots, n+1\}$ (ties broken at random), so $\Pr(V_{n+1} > V_{(r)}) \leq 1 - r/(n+1) \leq \alpha$; this is the split-conformal quantile lemma [3] applied to the scalar $V$. ∎

[1]The strict $\rho_i < c(\lambda)$ makes the set right-open; taking $c$ on a finite grid, as the timer's resolution of one inference does anyway, makes the minimum attained.

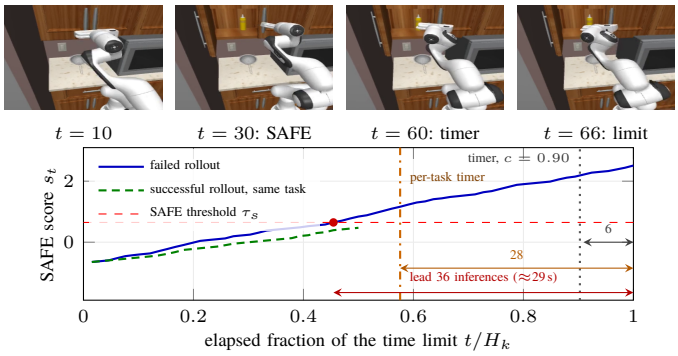



Fig. 1. One failed *OpenCabinet* rollout from the prospective test (chosen to illustrate; the median SAFE-first lead is smaller). The SAFE score crosses its calibrated threshold at $t = 30$ of $H_k = 66$ inferences, 36 inferences before the time limit; the timer alarms at $t = 60$, and the per-task timer, which never reads the policy, at $t = 38$. Dashed: the longest of the 9 (out of 10) successful test rollouts of this task that stay below the threshold; it ends at $t = 33$.

The path may use any data except calibration successes, such as development rollouts or calibration *failures*. The guarantee is marginal over tasks; a per-task version needs 19 calibration successes per task.

The preliminary results use the original protocol's joint search: among all pairs of calibration maxima with calibration FPR $\leq \alpha$, keep the one with the largest mean lead (highest recall for time-risk fusions). This best-of-many selection on 114 successes has no guarantee and is optimistic (test FPR 0.058–0.068 against 0.044 on calibration). The per-task timer rows, and the confirmatory test, use Proposition 1 with the path $\delta = \lambda$, $\tau_s = u + s\lambda$ and $(u, s)$ chosen on calibration failures.

## VI. Preliminary Evidence

**Setup.** We use the released Xiaomi-Robotics-1 RoboCasa365 checkpoint [25], which executes each 16-step action chunk in full, on the 38 official RoboCasa365 tasks [15, 16] with published success between 5 and 45 of 50 and enough failures (10 atomic, 28 composite; failures last 29–272 inferences). Disjoint cohorts serve one purpose each: development (1,824 rollouts) fits SAFE and the time risk; calibration (3 successes and 3 failures per task; 454 rollouts including those past the quota) sets thresholds; the prospective test (10 per class per task) is evaluated once; a confirmatory cohort of the same size is used only in Sec. VII. SAFE, its normalization and the time risk were frozen before the test was collected; the timers and fused detectors were added later, with thresholds from calibration only, so the results below are exploratory.

**The timer is the baseline to beat.** The timer ($c = 0.903$) alarms on all 380 test failures at 0.047 FPR with a fixed lead of 0.090 (Table I). The time-risk curve misses 140 failures, all in 14 tasks whose smoothed curve never reaches its threshold, so recall gains measured against it (to 0.90–0.96 for SAFE and four success-only scores) vanish here.

TABLE I

PROSPECTIVE TEST (380 SUCCESSES, 380 FAILURES), THRESHOLDS FROM CALIBRATION AT FPR $\leq 0.05$. $\bar{\ell}$: MEAN LEAD (MISS = 0); MED: MEDIAN LEAD (INFERENCES) OF DETECTED FAILURES; @.75: FAILURES ALARMED BY $0.75\,H_k$. *MEAN OF 20 DRAWS ($\bar{\ell}$ 0.088–0.134). †MARGIN BY PROPOSITION 1.

| Detector | Recall | FPR | $\bar{\ell}$ | Med | @.75 |
|---|---|---|---|---|---|
| Timer ($c = 0.903$) | 1.000 | 0.047 | 0.090 | 11.5 | 0.000 |
| Time risk | 0.632 | 0.047 | 0.120 | 16.5 | 0.211 |
| Time risk + timer | 1.000 | 0.050 | 0.125 | 11.0 | 0.105 |
| Noise + timer* | 1.000 | 0.066 | 0.101 | – | – |
| SAFE only | 0.566 | 0.039 | 0.190 | 23.0 | 0.421 |
| SAFE + time risk | 0.955 | 0.058 | 0.184 | 12.0 | 0.311 |
| Per-task timer† | 1.000 | 0.045 | 0.195 | 13.5 | 0.342 |
| SAFE + per-task timer† | 1.000 | 0.042 | 0.202 | 14.0 | 0.389 |
| **SAFE + timer** | 1.000 | 0.058 | 0.207 | 10.0 | 0.421 |

TABLE II

CONFIRMATORY TEST (380 SUCCESSES, 380 FAILURES). P = PER-TASK TIMER ($q = 0.95$). INTERVALS FROM P DOWN: REGISTERED DECISION BOOTSTRAP; ABOVE P: TEST ONLY. *MEAN OF 20 DRAWS [MAXIMUM]; RESAMPLED: SAFE, H&S. †FROZEN ORIGINAL THRESHOLDS. H&S: HIDE-AND-SEEK.

| Detector | Recall | FPR | $\bar{\ell}$ | Gain vs P |
|---|---|---|---|---|
| Global timer | 1.000 | 0.024 | 0.123 | $-0.102$ ($-0.147$, $-0.059$) |
| SAFE + time risk† | 0.955 | 0.039 | 0.190 | $-0.035$ ($-0.066$, $-0.007$) |
| SAFE + global timer | 1.000 | 0.047 | 0.215 | $-0.011$ ($-0.068$, 0.045) |
| P | 1.000 | 0.055 | 0.225 | – |
| Noise + P* | – | – | – | 0.002 [0.009] |
| Resampled + P* | – | – | – | 0.002, 0.000 [0.007] |
| **SAFE + P (H1)** | 1.000 | 0.061 | **0.255** | **0.030** (0.007, 0.093) |
| First-inference + P | 1.000 | 0.055 | 0.226 | 0.001 ($-0.015$, 0.036) |
| Zero-input + P | 1.000 | 0.061 | 0.237 | 0.012 (0.001, 0.044) |
| Shuffle + P | 1.000 | 0.063 | 0.236 | 0.011 (0.000, 0.064) |
| **H&S + P (Am. 2)** | 1.000 | 0.061 | **0.296** | **0.071** (0.022, 0.157) |

**SAFE adds lead over a single timer.** SAFE + timer raises the mean lead to 0.207 (paired gain 0.117, CI 0.060–0.178) and flags 42% of failures by $0.75\,H_k$ (Fig. 1), at 0.058 test FPR; noise fused with the timer reaches 0.101.

**A per-task timer removes the gain (H1, exploratory).** The per-task timer reaches $\bar{\ell} = 0.195$ at 0.045 FPR without reading the policy. SAFE + timer leads it by an insignificant 0.012 (CI $-0.044$ to 0.063). SAFE + per-task timer reaches 0.202, a gain of 0.007 (CI $-0.003$ to 0.019), far below the 0.05 margin of H1.

**A sequence detector does warn earlier (exploratory).** Fused with P under Proposition 1, Hide-and-Seek gains 0.098 over P (CI 0.037–0.158), and 0.083 (0.022–0.143) against P loosened to its own 21 test false alarms. In that matched comparison SAFE gains 0.008, ActProbe 0.009, and the best of 100 noise draws 0.016. The zero-input control matches Hide-and-Seek's ROC-AUC (0.888 against 0.878) and its recall when fused with the time risk (0.974 against 0.976), yet adds 0.001 over P at matched FPR, and the same-task shuffle $-0.004$: elapsed time explains separation, not earliness.

**No early warning, and mostly no early failure (exploratory).** No detector flags over 8% of failures by half the limit. Is timeout-defined failure then trivial by construction? One labeler, blind to alarms, marked the point of no return (after which the policy could not finish in time without help) on 48 confirmatory-cohort failures (2 per atomic, 1 per composite task): 38 (79%) were still recoverable at the limit (atomic 17/20, composite 21/28), most often slow or nearly successful. A second labeler agreed on 11 of 12 (Cohen's $\kappa = 0.80$). Failure here mostly means not finishing in time, which is why elapsed time is so informative; an early "failure" alarm mostly predicts slowness. On the 10 decided failures, P alarmed after the point of no return on all, SAFE + P at or before it on 2 of 3 atomic and 0 of 7 composite ones (an illustration).

**A second policy, offline (RLDX-1, exploratory).** On existing RLDX-1 data (20 tasks, 100/100 test rollouts), all failures ran to the time limit, P reaches $\bar{\ell} = 0.271$ against 0.032 for the global timer, and SAFE (LSTM head) + P gains 0.034 over P (CI 0.008–0.067) at the same FPR, all on atomic tasks; the fused path was chosen in sample and noise + P matched the gain in 2 of 20 draws at a higher FPR.

**The confound in a published benchmark (exploratory).** In FIPER's released rollouts [19], length alone separates failures with ROC-AUC 0.92–1.00 in four of five tasks. A timer calibrated like FIPER's thresholds exceeds its published mean accuracy (0.83 against 0.78) but alarms at 0.63 of the horizon against 0.30: FIPER's advantage is earliness, the comparison we argue for (published against computed numbers, not a paired re-run).

## VII. CONFIRMATORY TEST OF H1 AND H2

**Protocol.** A fresh test cohort (new seeds, 10 per class per task) was collected with SAFE frozen and preregistered for recall against the time risk. Before any timer or lead was computed on it, but with its recall and FPR known, an amendment made the per-task timer P ($q = 0.95$) the comparator and $\Delta\bar{\ell}$ the endpoint, with the rules of Sec. IV. A new calibration cohort of the same design (380 successes) sets the thresholds of P and SAFE + P by Proposition 1. The decision interval resamples both cohorts and refits thresholds in each of 2,000 replicates. Every test failure again ran to $H_k$.

**H1 fails for SAFE, but the gain is not zero.** SAFE + P leads P by 0.030 of the time limit (Table II). Its decision CI lies above zero and its test FPR (0.061, Wilson 95% upper bound 0.089) meets the FPR rules, but the point gain is below the 0.05 margin, so H1 fails.

**H2 is supported.** Noise and within-task resampled SAFE scores, fused with P and calibrated the same way, gain at most 0.009 over P in 20 draws each. What SAFE adds is policy evidence, not an artifact of fusing any score with a timer.

**Task duration carries most of SAFE's lead.** P alone leads the global timer by 0.102, and the original SAFE + time-risk design, which beat the time risk on recall here, gives 0.035 *less* lead than P. A per-task timer tuned on the test cohort to SAFE + P's FPR leaves SAFE a gain of 0.015 (CI $-0.007$ to 0.041). Descriptively, the four success-only scores fused with P gain 0.020–0.034, and all five scores gain more on the 10 atomic tasks (0.045–0.081) than on the 28 composite ones (0.008–0.027).

**Amendment 2: Hide-and-Seek passes H1 and H2.** It was registered before any Hide-and-Seek score was computed on these cohorts. The retrained detector reproduced its exploratory scores exactly, and the analysis reproduced SAFE + P before reporting. Hide-and-Seek + P leads P by 0.071 (decision CI 0.022–0.157) at 0.061 FPR (Wilson bound 0.089), meeting every rule; noise and resampled scores gain at most 0.012. The three controls gain 0.001–0.012. Hide-and-Seek beats the first-inference control by 0.070 (CI 0.014–0.153), so its warning develops during the rollout rather than reflecting a hard-looking scene, and beats the zero-input and shuffle controls by 0.059 and 0.060 (CIs above zero). Against SAFE + P its advantage, 0.041 (CI $-0.011$ to 0.103), is not significant. The decision bootstrap skews upward (SAFE: 0.007–0.093 around 0.030), so here the point gain, not the interval, carries the decision.

## VIII. Stage 2 Plan

**E2: Second policy (H3).** RLDX-1 on its 20 tasks (Sec. VI), fresh calibration and test cohorts of 20 successes and 20 failures per task (about 2,500 rollouts), with held-out calibration failures. Co-primary: SAFE + P and Hide-and-Seek + P, each against P, superiority (decision 97.5% CI above zero), conditional on these 20 tasks (rollouts resampled within task), under the FPR rule below. Declared secondary: Hide-and-Seek + P against the first-inference control + P (Sec. V), whose CI above zero would show the warning develops during the rollout. At 95%, offline sizing gave SAFE power 0.92 at its offline gain of 0.034 (the CI excluded zero in every simulated study, so the FPR rule decided); a 0.05 margin would have power $\leq 0.09$, so we drop it.[PENDING: power at 97.5% for both scores, from the offline RLDX-1 Hide-and-Seek check] Also secondary, with no power claim: each co-primary's CI with tasks resampled (SAFE: power about 0.54 at 95%). Descriptive stratum: the 14 atomic tasks. It replaces an atomic-only XR-1 test, with at most 0.55 power at the 0.05 margin up to 60 per class per task and an optimistic 0.089 estimate (about 0.06 with all retained successes). With 5 held-out failures per task, all sizing figures are optimistic.

**FPR rule.**[TODO: Proposed rule; author to confirm.] Calibration keeps the conformal target $\alpha = 0.05$. H1's fixed limit (FPR $\leq 0.075$, Wilson bound $\leq 0.10$) rejects good detectors in small cohorts; instead, false alarms must not reject "true FPR $\leq 0.075$" in a one-sided exact binomial test at level 0.05: at most the 95th percentile of Binomial($n_s$, 0.075) for $n_s$ test successes (39 of 400 in E2). This is looser, so E2's power stays $\geq 0.92$.

**E3: Reset and retry (H4).** On XR-1's 38 tasks[PENDING: E3 policy not chosen; XR-1 assumed (RLDX-1: about 480 per arm)], an alarm ends the attempt and resets the scene while a budget of $1.25\,H_k$ remains. Registered: P against no monitor, paired by seed (about 650 episodes per arm). SAFE + P is descriptive: separating it from P needs thousands to over a million episodes per arm (about 1,450 on XR-1 atomic tasks at $1.0\,H_k$, on an optimistic bound). As most failures are still recoverable at the limit, a "more time, no reset" arm (run to $1.25\,H_k$) is the natural comparator for reset.[PENDING: sample size of the no-reset arm]

**E4: Checks.** Action-only monitors (chunk magnitude, second-sample disagreement); richer time baselines (end-effector stall, discrete-time hazard); OR vs AND on full curves; SAFE faithfulness; precision at deployment prevalence; unseen tasks; FIPER's code re-run with the timer. **E5: Failures decided early.** Most failures have no early point of no return, so E5 targets the minority that do, with several labelers, a written protocol and adjudication (our two labelers' timings differed by 0.24–0.34 $H_k$), and reports lead from that moment.

## IX. Limitations

All results are simulated shadow evaluations (monitors never intervened), from one policy prospectively; a second (RLDX-1) was checked only offline, on existing data with a small test set. Timers and fused detectors were defined after the exploratory cohort was opened, and the confirmatory endpoint after that cohort's recall and FPR were known. Hide-and-Seek was chosen as the better of two supervised baselines on the opened cohort, and registered after SAFE's confirmatory result was known, on cohorts already opened for SAFE. Part of SAFE + P's gain is its higher test FPR (0.061 against 0.055). Timers and normalization assume a known task and limit; the policy score needs white-box access. Test cohorts are balanced by quota, so recall and FPR are per class, not at deployment prevalence. Point-of-no-return labels are one labeler's video judgments, checked on 12 videos; only 10 failures were decided.

## X. Conclusion

When failure means timeout, and mostly slowness, a learned monitor can only warn earlier than a timer at the same false-alarm rate, here and on a published benchmark. Elapsed time explains how well monitors separate failures. On fresh preregistered cohorts, SAFE's hidden-state score added 0.030 of the time limit to a per-task timer, below our margin, while a sequence detector trained for early separation on the same features added 0.071, a warning that develops during the rollout. Stage 2 tests a second policy and closed-loop retry.

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